



B. Network Disentanglement

CPU: 3.00s

Memory: 2048MB

Though Maryland is a remote area, luckily there are broadband internet connection services. Maryland is presented as a $2D$ geometric plane, where there exist N routers and M wire connections (straight line segments) that link pairs of routers located on the $2D$ plane. There are only two internet service providers, X and Y . As these connections are placed on a $2D$ geometric plane, it can happen that any two wires cross or touch each other. As a result, they face problems while providing service when wires of different ISPs cross or touch each other (they can handle when their own wires cross or touch each other). Now, they need your help to solve their problem. Your task is to find the minimum number of wires to remove such that no two wires of different ISPs can cross or touch each other, and you also have to find which wires to remove.

NB: Two wires cross or touch each other when the intersecting point(s) lies on both wires. So it can happen that the two wires lie on the same line. There can be multiple wires between any two routers. There won't be a wire with both endpoints connected to the same router.

Input Description

The first line will contain $T (1 \leq T \leq 50)$, the number of test cases. Each case will start with two integers $N (2 \leq N \leq 1000)$, $M (1 \leq M \leq 1000)$, where N denotes the number of routers and M represents the number of wire connections in Maryland. Each of the next N lines will contain two integers: $X_i (-10^9 \leq X_i \leq 10^9)$ and $Y_i (-10^9 \leq Y_i \leq 10^9)$, denoting the location of the i^{th} router. No two routers are situated in the same place. Each of the next M lines will contain three integers: $U_i (1 \leq U_i \leq N)$, $V_i (1 \leq V_i \leq N)$, and $I (1 \leq I \leq 2)$, where $U_i \neq V_i$, denoting the i^{th} wire link between routers U_i and V_i in the $2D$ plane, and I denotes whose wire it is ($I = 1$ for X , $I = 2$ for Y).

Output Description

For each case, you need to print "**Case #T:**" in the first line (without quotes), where T is the test case number. The second line contains an integer denoting the minimum number of wires to remove such that no two wires of different ISPs can cross or touch each other. The third line of each case contains space-separated wire link numbers (1-based) to remove. **The third line will be an empty line if you don't have to remove a wire link.**

Samples

Input

```
2
4 2
-1 -1
-1 1
1 -1
1 1
1 2 1
3 4 1
6 3
-2 0
2 0
-1 -1
-1 1
1 -1
1 1
1 2 1
3 4 2
5 6 2
```

Output

```
Case #1:
0

Case #2:
1
1
```

ICPC PRELIMINARY DHAKA SITE 2024

Ends in
02:49:49

THE CONTEST IS RUNNING

SUBMIT

Language

C++

-O2 -static -std=gnu++17



SELECT SOURCE

SUBMIT

MY SUBMISSIONS



No submissions yet!